



PTT PHENOL COMPANY
LIMITED

PROCESS SPECIFICATION

PREFLASH AND FLASH COLUMNS OVERHEAD PUMPS (P-2307A/B)

Docu. No. : 14780-8120-PS-P2307
Project No. : 120117
Rev. No. : Z1
Date : MAY 10, 2016
Page : 1 / 6

TITLE :

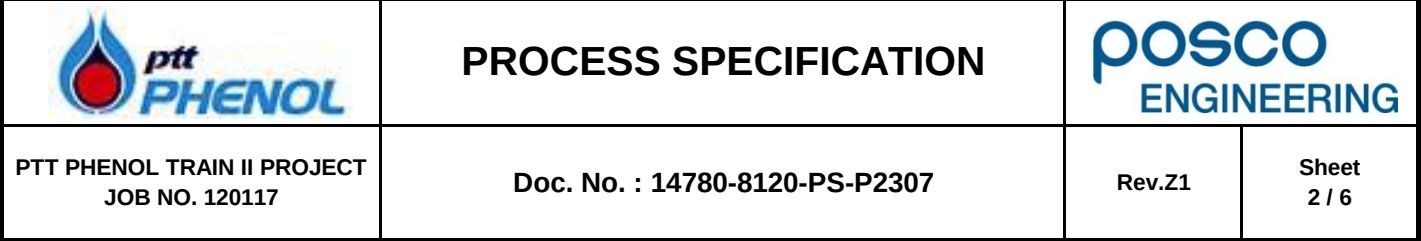
PREFLASH AND FLASH COLUMNS OVERHEAD PUMPS

(P-2307A/B)

DISCIPLINE : PROCESS
DOCUMENT NO. : 14780-8120-PS-P2307
PROJECT NAME : PTT Phenol Train II Project
PROJECT NO. : 120117
OWNER : PTT Phenol Company Limited (PPCL)

Z1	MAY 10, 2016	AS BUILT	S.W.Park	J.S.Chung	-	S.W.Lee
F1	MAY 23, 2014	FINAL ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
A2	FEB 25, 2014	APPROVED FOR DESIGN	S.W.Park	J.S.Chung	-	S.W.Lee
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REV. NO.	DATE	DESCRIPTION	PREP'N	REVIEW	QA REVIEW	APPROVAL
REV. NO.	PREPARATION DEPT.	PREPARATION	REVIEW	QA REVIEW	APPROVAL	
Z1	PROCESS	S.W.Park	J.S.Chung	-	S.W.Lee	
		MAY 10, 2016	MAY 10, 2016	-	MAY 10, 2016	

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Record of Revision	
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				PROJECT SPECIFICATION																																			
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CENTRIFUGAL PUMPS AND DRIVERS				<table border="1" style="width:100%; border-collapse: collapse;"> <tr> <th>REV</th> <th>DATE</th> <th>BY</th> <th>APVD</th> <th>REV</th> <th>DATE</th> <th>BY</th> <th>APVD</th> </tr> <tr> <td>0</td> <td>09-Apr-12</td> <td>MW</td> <td>DRL</td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>1</td> <td>25-Apr-12</td> <td>MW</td> <td>MEH</td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td> </td> <td> </td> <td> </td> <td> </td> <td> </td> <td> </td> <td> </td> <td> </td> </tr> </table>				REV	DATE	BY	APVD	REV	DATE	BY	APVD	0	09-Apr-12	MW	DRL					1	25-Apr-12	MW	MEH												
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APPLICABLE TO: ☐ PROPOSALS ☐ PURCHASE ☒ AS BUILT				1																																			
FOR _____ UNIT _____				2																																			
SITE _____ SERVICE Preflash and Flash Columns Overhead Pumps				3																																			
NO. REQ 2 PUMP SIZE _____ TYPE Process NO. STAGES 1				4																																			
MANUFACTURER _____ MODEL _____ SERIAL NO. _____				5																																			
NOTES: INFORMATION BELOW TO BE COMPLETED: ☐ BY PURCHASER ☐ BY MANUFACTURER ☒ BY MANUFACTURER OR PURCHASER				6																																			
GENERAL				7																																			
PUMPS TO OPERATE IN (PARALLEL) _____ NO. MOTOR DRIVEN 2 NO. TURBINE DRIVEN _____				8																																			
(SERIES) WITH _____ PUMP ITEM NO. P-2307A/B PUMP ITEM NO. _____				9																																			
GEAR ITEM NO. _____ MOTOR ITEM NO. _____ TURBINE ITEM NO. _____				10																																			
GEAR PROVIDED BY _____ MOTOR PROVIDED BY _____ TURBINE PROVIDED BY _____				11																																			
GEAR MOUNTED BY _____ MOTOR MOUNTED BY _____ TURBINE MOUNTED BY _____				12																																			
GEAR DATA SHT. NO. _____ MOTOR DATA SHT. NO. _____ TURBINE DATA SHT. NO. _____				13																																			
OPERATING CONDITIONS (5.1.3)				SITE AND UTILITY DATA (CONT'D)																																			
☐ CAPACITY, NORMAL 211 m³/h RATED 232 m³/h				14																																			
☐ OTHER _____ m³/h				15																																			
☐ SUCT PRESS MAX./RATED 4.167 / -.4321 kg/cm²[g]				16																																			
☐ DISCHARGE PRESSURE 7.073 kg/cm²[g]				17																																			
☐ DIFFERENTIAL PRESSURE 7.50 kg/cm²				18																																			
☐ DIFF. HEAD 88.6 m NPSHA 6.9 m				19																																			
☐ PROCESS VARIATIONS _____				20																																			
☐ STARTING CONDITIONS _____				21																																			
SERVICE: ☒ CONT. ☐ INTERMITTENT (STARTS/DAY) _____				22																																			
☐ PARALLEL OPERATION REQ'D				23																																			
SITE AND UTILITY DATA				24																																			
LOCATION:				25																																			
☐ INDOOR ☐ HEATED ☐ UNDER ROOF				26																																			
☒ OUTDOOR ☒ UNHEATED ☐ PARTIAL SIDES				27																																			
☒ GRADE ☐ MEZZANINE _____				28																																			
ELECTRIC AREA CLASSIFICATION				29																																			
CL 1 GR IIA DIV ZONE 2				30																																			
☐ WINTERIZATION REQ'D ☒ TROPICALIZATION REQ'D.				31																																			
SEE NOTES ON SHEET 6 *				32																																			
SITE DATA				PERFORMANCE																																			
ALTITUDE _____ BAROMETER _____				33																																			
RANGE OF AMBIENT TEMPS: MIN/MAX. _____ / _____				34																																			
RELATIVE HUMIDITY: MIN / MAX _____ / _____ %				35																																			
UNUSUAL CONDITIONS: ☐ DUST ☐ FUMES				36																																			
OTHER _____				37																																			
UTILITY CONDITIONS:				38																																			
STEAM: DRIVERS HEATING				39																																			
MIN _____				40																																			
MAX _____				41																																			
ELECTRICITY DRIVERS HEATING CONTROL SHUTDOWN				42																																			
VOLTAGE _____				43																																			
HERTZ _____				44																																			
PHASE _____				45																																			
COOLING WATER:				46																																			
TEMP. INLET _____ MAX. RETURN _____				47																																			
PRESS. NORM _____ DESIGN _____				48																																			
MIN RETURN _____ MAX ALLOW DP _____				49																																			
FOULING FACTOR _____				50																																			
PROPOSAL CURVE NO. _____ RPM 2950				51																																			
☐ IMPELLER DIA. RATED _____ MAX. _____ MIN. _____ mm				52																																			
☐ RATED POWER 65.0 kW EFFICIENCY 73.0 %				53																																			
☐ MINIMUM CONTINUOUS FLOW:				54																																			
THERMAL _____ m³/h STABLE _____ m³/h				55																																			
☐ PREFERRED OPERATING REGION _____ TO _____ m³/h				56																																			
☐ ALLOWABLE OPERATING REGION _____ TO _____ m³/h				57																																			
☐ MAX HEAD @ RATED IMPELLER _____ m				58																																			
☐ MAX POWER @ RATED IMPELLER _____ kW				59																																			
☐ NPSHR AT RATED CAPACITY _____ m				60																																			
☒ SUCTION SPECIFIC SPEED _____				61																																			
☐ MAX. SOUND PRESS. LEVEL REQ'D _____ (dBA)				62																																			
☐ EST MAX SOUND PRESS. LEVEL _____ (dBA)				63																																			
REMARKS: See Pump Notes 2, 6, C1, C2 and 42. See Motor Note M10.				64																																			
ESTIMATED SHUTOFF PRESSURE 13.55 kg/cm²[g]				65																																			

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CENTRIFUGAL PUMPS AND DRIVERS

PROJECT SPECIFICATION

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REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	09-Apr-12	MW	DRL				

REV	CONSTRUCTION	CONSTRUCTION (CONT)																						
	APPLICABLE STANDARD: ☒ API 610 11TH EDITION ☐ OTHER _____ (SEE REMARKS)	☐ SHAFT DIAMETER BETWEEN BEARINGS _____ mm ☐ SPAN BETWEEN BEARING CENTERS _____ mm ☐ SPAN BETWEEN BEARING & IMPELLER _____ mm	1																					
	PUMP TYPE: ☒ OH2 ☐ BB1 ☐ VS1 ☐ VS6 ☐ OH3 ☐ BB2 ☐ VS2 ☐ VS7 ☐ OH4 ☐ BB3 ☐ VS3 ☐ OTHER ☐ OH5 ☐ BB4 ☐ VS4 ☐ OH6 ☐ BB5 ☐ VS5	REMARKS: COUPLINGS: _____ DRIVER - PUMP ☐ MAKE _____ ☒ MODEL _____ ☐ RATING _____ ☐ LUBRICATION _____ ☒ LIMITED END FLOAT REQUIRED _____ ☒ SPACER LENGTH _____ mm ☒ SERVICE FACTOR _____	2																					
	NOZZLE CONDITIONS: <table border="1" style="width:100%; border-collapse: collapse;"> <thead> <tr> <th>SIZE (IN)</th> <th>FLANGE CLASS</th> <th>FAC'G</th> <th>POSITION</th> </tr> </thead> <tbody> <tr> <td>SUCTION</td> <td>300</td> <td>RF</td> <td></td> </tr> <tr> <td>DISCHARGE</td> <td>300</td> <td>RF</td> <td></td> </tr> <tr> <td>BALANCE DRUM</td> <td></td> <td></td> <td></td> </tr> </tbody> </table>	SIZE (IN)	FLANGE CLASS	FAC'G	POSITION	SUCTION	300	RF		DISCHARGE	300	RF		BALANCE DRUM				DRIVER HALF COUPLING MOUNTED BY: ☐ PUMP MFR. ☐ DRIVER MFR. ☐ PURCHASER ☐ COUPLING PER API 671	3					
SIZE (IN)	FLANGE CLASS	FAC'G	POSITION																					
SUCTION	300	RF																						
DISCHARGE	300	RF																						
BALANCE DRUM																								
	PRESSURE CASING CONNECTIONS: <table border="1" style="width:100%; border-collapse: collapse;"> <thead> <tr> <th>NO.</th> <th>SIZE (NPS)</th> <th>TYPE</th> </tr> </thead> <tbody> <tr> <td>☐ DRAIN</td> <td></td> <td>Provide With Valve</td> </tr> <tr> <td>☐ VENT</td> <td></td> <td></td> </tr> <tr> <td>☐ PRESSURE GAUGE</td> <td></td> <td></td> </tr> <tr> <td>☐ TEMP GAUGE</td> <td></td> <td></td> </tr> <tr> <td>☐ WARM-UP</td> <td></td> <td></td> </tr> <tr> <td>☐ BALANCE / LEAK-OFF</td> <td></td> <td></td> </tr> </tbody> </table>	NO.	SIZE (NPS)	TYPE	☐ DRAIN		Provide With Valve	☐ VENT			☐ PRESSURE GAUGE			☐ TEMP GAUGE			☐ WARM-UP			☐ BALANCE / LEAK-OFF			BASEPLATES: API BASEPLATE NUMBER _____ (APPENDIX M) ☐ NON-GROUT CONSTRUCTION:	4
NO.	SIZE (NPS)	TYPE																						
☐ DRAIN		Provide With Valve																						
☐ VENT																								
☐ PRESSURE GAUGE																								
☐ TEMP GAUGE																								
☐ WARM-UP																								
☐ BALANCE / LEAK-OFF																								
	☐ CYLINDRICAL THREADS REQUIRED CASING MOUNTING: (SEE SEPARATE SHEET FOR VERTICALS) ☒ CENTERLINE ☐ NEAR CENTERLINE ☐ FOOT ☐ SEPARATE MOUNTING PLATE ☐ IN-LINE	MATERIAL ☐ APPENDIX H CLASS _____ A-8 ☐ MIN DESIGN METAL TEMP _____ 15 °C ☐ BARREL/CASE 316 SS IMPELLER 316 SS ☐ CASE/IMPELLER WEAR RINGS Hard-Faced 316 SS ☐ SHAFT 316 SS ☐ DIFFUSERS _____ ☐ COUPLING SPACER/HUBS _____ ☐ COUPLING DIAPHRAGMS (DISKS) _____	5																					
	CASING SPLIT: ☐ AXIAL ☒ RADIAL		6																					
	CASING TYPE: ☐ SINGLE VOLUTE ☒ MULTIPLE VOLUTE ☐ DIFFUSER ☒ OVERHUNG ☐ BETWEEN BEARINGS ☐ BARREL		7																					
	CASE PRESSURE RATING: ☐ MAX ALLOWABLE WORKING PRESSURE _____ kg/cm ² [g] @ _____ 0 ☐ HYDROTEST PRESSURE _____ kg/cm ² [g] ☐ SUCT'N PRESS. REGIONS MUST BE DESIGNED FOR MAWP	BEARINGS AND LUBRICATION BEARING (TYPE/NUMBER): ☐ RADIAL _____ Ball / _____ ☐ THRUST _____ Ball / _____ ☐ REVIEW AND APPROVE THRUST BEARING SIZE LUBRICATION: ☒ GREASE ☒ FLOOD ☐ RING OIL ☒ FLINGER ☐ PURGE OIL MIST ☒ PURE OIL MIST ☐ CONSTANT LEVEL OILER PREFERENCE (SEE REMARKS) ☐ PRESSURE LUBE SYS ☐ API-610 ☐ API-614 ☒ OIL VISC. ISO GRADE _____	8																					
	ROTATION: (VIEWED FROM COUPLING END) ☐ CW ☐ CCW ☐ IMPELLERS INDIVIDUALLY SECURED		9																					
	REMARKS: ☐ BOLT OH3 PUMP TO PAD/FOUNDATION SHAFT: ☐ SHAFT DIAMETER AT COUPLING _____ mm		10																					
	REMARKS: SUCTION TYPE <u>Single</u> IMPELLER TYPE <u>Closed</u> Pumps shall comply with UOP Standard Specifications 5-11 and 7-12 See Project Specification 533.		11																					

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CENTRIFUGAL PUMPS AND DRIVERS

PROJECT SPECIFICATION

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REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	09-Apr-12	MW	DRL				

REV	BEARINGS AND LUBRICATION (CONT)	MECHANICAL SEAL OR PACKING (CONT)	
	☒ OIL HEATER REQ'D ☐ ELECTRIC ☐ STEAM ☐ OIL PRESS TO BE GREATER THAN COOLANT PRESS REMARKS 	☐ VAPOR PRESSURE _____ kg/cm ² [a] @ _____ °C ☐ HAZARDOUS ☐ FLAMMABLE ☐ OTHER _____ ☐ FLOW RATE MAX/MIN _____ / _____ m ³ /h ☐ PRESSURE REQUIRED MAX/MIN _____ / _____ kg/cm ² [g] ☒ TEMPERATURE REQUIRED MAX/MIN _____ / _____ °C	1
		QUENCH FLUID: ☐ NAME OF FLUID _____ ☐ FLOW RATE _____ m ³ /h SEAL FLUSH PIPING: ☐ SEAL FLUSH PIPING PLAN _____ API Plan 11 ☒ TUBING ☐ CARBON STEEL ☐ PIPE ☒ STAINLESS STEEL ☐ AUXILIARY FLUSH PLAN _____ API PLAN 53A ☒ TUBING ☐ CARBON STEEL ☐ PIPE ☒ STAINLESS STEEL ☐ PIPING ASSEMBLY: ☒ THREADED ☒ UNIONS ☐ SOCKET WELDED ☐ FLANGED ☒ TUBE TYPE FITTINGS ☒ LOW PRESSURE SWITCH (PLAN 52/53) TYPE _____ ☒ PRESSURE GAUGE (PLAN 52/53) ☒ LOW LEVEL SWITCH (PLAN 52/53) TYPE _____ ☒ LEVEL GAUGE (PLAN 52/53) ☐ TEMP INDICATOR (PLANS 21, 22, 23, 32, 41) ☒ HEAT EXCHANGER (PLAN 52/53)	2
		REMARKS 	3
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0	09-Apr-12	MW	DRL				
1	13-Aug-12	MW	DRL				

REV

NOTES:

* SEE BASIC ENGINEERING DESIGN QUESTIONNAIRE (BEDQ), ENGINEERING DESIGN INFORMATION (EDI) PROJECT 963763 A.4, SECTIONS 3 AND 4.

The following notes apply to those services for which they are specified in this Project Specification:

PUMP NOTES:

2. Pump vendor shall provide non-contacting pressurized dry gas seals. The barrier gas shall be Nitrogen from the plant Nitrogen header available at 7 kg/cm²(g). Pump vendor shall provide a seal control panel for each pump. The panel shall include the following:

- Coalescing filter (removal efficiency of 98% on liquids 1 micron or greater and solids 3 microns or greater)
- Pressure regulating valve
- Pressure gauge
- Flow meter low range
- Flow meter high range

Alarm switches:

- Low Pressure
- High Flow

The alarm switches shall activate a control center trouble alarm (by others). The contractor and pump vendor shall coordinate switch selection.

A mechanical amplifier shall be provided by the seal vendor to boost the nitrogen buffer supply pressure when the normal nitrogen supply pressure does not exceed either:

- the pump suction pressure by 2.1 kg/cm², or
- the pump maximum suction pressure by 0.35 kg/cm² psi.

When these conditions exist, the seal gas control panel scope shall be expanded to include the amplifier and a minimum 2 gallon stainless steel accumulator vessel with local pressure gauge.

6. Mechanical seal parts and packings shall be suitable for handling aromatic hydrocarbons.

9. The "capacity, other" is based on 50% of the lowest normal operating flow.

11. Pumps shall be mounted on cover plates to be furnished by vessel supplier to suit sump tank opening. Each pump shall include a stainless steel suction screen. A steel discharge pipe shall extend through the cover plate terminating in a Cl. 300 flanged connection. Contractor to determine depth of setting. Pump will be spared by common warehouse spare pump P-1304S.

42. VOC packing valves should be supplied.

C1. Casing volume to be minimized and subjected to critical volume restriction

C2. Cumene Hydroperoxide is thermally unstable that can initiate self-heating reaction at elevated temperatures

C3. Acid Aromatics sump pump P-2304A shall be identical to the existing Acid Aromatics sump pump in PPCL Unit 1 (Item No. P-1304A/S). Existing warehouse spare pump P-1304S shall be used as a common spare for both services.

MOTOR NOTES:

M5. Non Overloading at all points on pump curve

M10. Motor shall have torque capability to accelerate the pump to operating speed with discharge shutoff valve open, and the discharge check valve closed against a static pressure of 1.8 kg/cm² (g)

M12. Deleted.

M13. The standby pump motor shall be connected to an alternate independent power supply.

GENERAL NOTES:

G1. Lubrication system is applied to high voltage motor's and all pumps' bearings except for vertical and diaphragm pumps.

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